ABU NAYEM

Senior Backend & AI Engineer · Software Architect & Tech Entrepreneur

Dhaka-1217, Bangladesh | **Phone / WhatsApp:** +880 1991-953820 | **Email:** naeem043@gmail.com
Portfolio: https://naeem043.github.io | **GitHub:** github.com/naeem043

PROFESSIONAL SUMMARY

Results-driven **Senior Backend & AI Engineer and Tech Lead with 8+ years of experience** architecting, building, and scaling production backend systems, intelligent AI-powered applications, multi-tenant SaaS products, and enterprise ERP/HRM platforms. Deep specialization in **Python, Django, Django REST Framework (DRF), and FastAPI**, combined with hands-on expertise in **Large Language Models (LLMs), Retrieval-Augmented Generation (RAG & LightRAG), LangChain, Microsoft Agent Framework, MCP Servers, Vector Databases, and n8n workflow automation**. Proven track record of leading distributed engineering teams across the USA, UK, and Bangladesh, optimizing high-throughput relational databases, managing Linux/Docker/cloud infrastructure, and turning complex business requirements into reliable, revenue-generating software.

TECHNICAL SKILLS & TOOLING

Backend Engineering (Primary)	Python, Django, Django REST Framework (DRF), FastAPI, RESTful APIs, Microservices & System Architecture, PayloadCMS, Laravel
AI, LLMs & Intelligent Apps	LLMs (OpenAI, Claude), LangChain, Microsoft Agent Framework, RAG, LightRAG, Vector Databases, Embeddings & Semantic Search, AI Chatbots, RASA, MCP Server, n8n, OpenCV, TensorFlow
Databases & Performance	PostgreSQL, MySQL, MS SQL Server, SQLite, Schema Design, Indexing & Query Optimization, High-Volume Data Processing, Redis
DevOps & Cloud Infrastructure	Linux Server Administration, Docker, Nginx, Gunicorn, Daphne, CI/CD Pipelines, Celery, Redis, Cloud Deployment & Server Security
Frontend & Mobile Development	Next.js, React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap, jQuery, AJAX, Flutter (Cross-Platform Mobile)
Third-Party Integrations	PayPal, Stripe, SSLCommerz, bKash, Microsoft Graph API, Google Cloud Console APIs, Typesense, Zoom, Zoho, WhatsApp Cloud API, Facebook Messenger API, SMS Gateways, Webhooks
Authentication & Identity (SSO)	Auth0, Okta, Keycloak, Microsoft Entra ID, Google Workspace, AWS Cognito, OAuth 2.0, OpenID Connect (OIDC), SAML SSO, JWT
IDEs, AI Dev Tools & Workflow	VS Code, Cursor, Claude Code, Claude CLI, Antigravity, GitHub Copilot, Git, GitHub, Postman, Jira, Slack, Microsoft Teams

PROFESSIONAL EXPERIENCE

Senior Backend & AI Engineer *(Remote)* **2023 — Present**

G Group *— Florida (FL), USA*

- Architect, develop, and maintain scalable backend systems, high-performance REST APIs, and multi-tenant SaaS products using Python, Django, Django REST Framework (DRF), and FastAPI.
- Design and deploy intelligent AI applications powered by Large Language Models (LLMs), Retrieval-Augmented Generation (RAG & LightRAG), LangChain, AI Agents, and custom Model Context Protocol (MCP) servers.
- Build conversational AI chatbots and end-to-end n8n workflow automations integrating third-party SaaS platforms, payment gateways, webhooks, and semantic vector search pipelines.
- Lead relational and vector database schema design, indexing strategies, and query optimization for large-scale production datasets.
- Provide technical leadership, architectural reviews, and cross-functional collaboration with product, frontend, and DevOps teams to deliver real-world business solutions.

  Tech Stack: *Python, Django, DRF, FastAPI, LLMs, LangChain, Microsoft Agent Framework, RAG, LightRAG, Vector DBs, MCP Server, n8n, PostgreSQL, Redis, Celery, Docker, Linux*

Sr. Software Engineer & Team Lead *(Part-time · Remote)* **Aug 2021 — July 2023**

Av Tech Ltd. *— London, UK*

- Led the Bangladesh-based development team — personally building and enhancing Django/DRF features while leading the Laravel, React, and Next.js project members.
- Managed project execution, daily team stand-ups, task monitoring, testing, and technical recruitment of team members.
- Directed Git management, code reviews, Linux server & DevOps administration, and security & performance analysis at all levels.
- Conducted regular management meetings with the company owner to align engineering delivery with business goals.

  Tech Stack: *Django, DRF, Laravel, React, Next.js, PostgreSQL, MySQL, Tailwind CSS, Celery, Redis, Nginx, Gunicorn*

Backend Developer *(Part-time · Remote)* **Jun 2020 — Nov 2021**

Fast Despatch Logistics Ltd. (FDL) *— Peterborough, UK*

- Served as the main developer responsible for three projects — a Django REST API platform and two Django-rendered web applications, including frontend implementation.
- Collaborated with the Project Manager on technical requirements analysis, built new features and custom reports, and handled server management.
- Authored API documentation, led integration testing, guided frontend and mobile app developers, and provided production support & issue resolution.

  Tech Stack: *Django, DRF, HTML, CSS, JavaScript, Bootstrap*

Sr. Software Engineer & Team Lead **Dec 2020 — Jul 2023**

Esquire Technology Ltd. *— Tejgaon, Dhaka*

- Supervised and led a team of 3 software engineers building the company's in-house enterprise ERP system to global standards.
- Owned project planning & management, database & system design, code/requirements/risk analysis, and converting business requirements into design documents.
- Integrated custom software with existing enterprise systems and performed root-cause analysis, code reviews, and unit testing.
- Awarded 'Software Engineer of the Month' 3 times (2021–2022) in recognition of outstanding technical leadership and delivery.

  Tech Stack: *Django, DRF, PostgreSQL, Celery, Redis, Nginx, Gunicorn, Bootstrap*

Software Engineer *(Remote)* **Feb 2020 — Nov 2020**

ISHO Limited *— Banani, Dhaka*

- Worked with the ERP and e-commerce teams, building the company ERP and integrating it with the e-commerce platform.
- Conducted requirements analysis with the business team, designed flowcharts & process-flow diagrams, and delivered user training & end-user support.

  Tech Stack: *Django, MySQL, HTML, CSS, JavaScript, Bootstrap*

Software Developer **Jan 2017 — Jun 2020**

Pythonbd IT Solution *— Agargaon, Dhaka*

- Built custom software, multi-branch POS & inventory software, Education ERP systems, and client websites & web applications to client requirements.
- Performed requirements analysis, collaborated with other developers, trained agents to implement the Education ERP in institutions, and provided ongoing project support.

  Tech Stack: *Django, MySQL, PostgreSQL, SQLite, Bootstrap*

KEY PROJECTS & PRODUCT ARCHITECTURE

Enterprise RAG & Knowledge Intelligence Platform | *AI / RAG & Knowledge System* *Role: RAG Pipeline Architecture, Backend API & AI Integration*

Architected a Retrieval-Augmented Generation (RAG) platform that ingests enterprise documents and internal data sources, indexes them with vector embeddings, and delivers accurate, context-grounded answers with source citations through a conversational interface and MCP server tools. **[Python, FastAPI, LangChain, LLM, RAG, Vector DB, Embeddings, Semantic Search, PostgreSQL, Next.js]**

Business Automation Chatbot | *AI Chatbot / Omnichannel Automation* *Role: Backend Development, n8n Orchestration & Multi-Channel Integration*

Built an omnichannel AI chatbot and workflow automation platform automating lead capture and qualification, instant Facebook Messenger and WhatsApp replies, automated email follow-ups, and end-to-end customer communication pipelines. **[n8n, Python, Django, DRF, WhatsApp Integration, Facebook Messenger API, Email Service, Webhooks, LLM, PostgreSQL]**

Construction ERP | *ERP / Construction & Supply Chain* *Role: System Architecture, Database Design & Full-Stack Development*

Delivered an end-to-end ERP for construction and engineering firms covering site material requisitions (MR), purchase orders (PO), multi-warehouse inventory & stock tracking, supply chain & vendor management, project BOQ & accounting ledgers, HRM, role-based access control (RBAC), and automated PDF & Excel reporting. **[Python, Django, DRF, PostgreSQL, Supply Chain & Inventory, RBAC, Accounts & HRM, PDF & Excel Reports]**

Fund Management ERP | *ERP / Financial & Employee Funds* *Role: Database Design & Full-Stack Development*

Built a specialized financial and employee fund management ERP for managing Provident Fund (PF), Gratuity Fund, Workers' Welfare Fund, member contributions, loan & settlement workflows, interest distribution, accounting ledgers, and automated PDF & Excel reporting. **[Python, Django, PostgreSQL, HTML, CSS, Bootstrap, jQuery, Provident & Gratuity Fund, PDF & Excel Reports]**

Quotation Management Software | *B2B SaaS / Sales & Billing* *Role: Backend Architecture & Full-Stack Development*

Engineered a commercial quotation and billing lifecycle system featuring role-based user management, quotation creation, drafting, submission & multi-revision tracking, quotation-linked sales and milestone billing realization, and one-click PDF & Excel reporting. **[Python, Django, PostgreSQL, Quotation Workflow, Sales & Billing Tracking, Role-Based Access, PDF & Excel Export]**

Employee Monitoring Tracker Desktop App | *Desktop App / HR & Workforce Analytics* *Role: Python Desktop App, Django REST API Backend & Admin Portal*

Built a cross-platform Python desktop time and activity tracker integrated with a Django REST Framework backend and a Django-based web admin interface to manage desktop clients, monitor employee screen captures and active hours, and automate payroll billing, overtime bonuses, and late/idle fine calculations. **[Python Desktop App, Django, DRF, PostgreSQL, Time & Activity Tracking, Billing/Bonus/Fine Engine, Admin Portal]**

Garments ERP | *ERP / Textile & Garments* *Role: Project Planning, Database & System Design, Backend Development*

Designed and built a fully integrated ERP managing the complete textile-to-garments lifecycle — from raw material procurement, BOM costing, and production floor tracking through to finished goods and export dispatch. **[Python, Django, PostgreSQL, Celery, Redis]**

Mobile Servicing ERP — Grameen Distribution Ltd. | *ERP / After-Sales Service* *Role: Backend Development & Database Design*

Developed a custom high-volume after-sales ERP managing all nationwide mobile care centers under Grameen Distribution Ltd., engineered to process thousands of daily service transactions, warranty claims, and spare-parts movements. **[Python, Django, PostgreSQL, High-Volume Data Processing]**

Education ERP (School Management System) | *ERP / Education* *Role: Full-Stack Development & Implementation Training*

Built a large-scale institutional ERP covering executive analytics dashboards, student admission & academic modules, automated GPA/CGPA result tabulation & marksheets, online/offline fee payment, and QR-verified admit card generation across multiple institutions. **[Python, Django, MySQL, Bootstrap]**

HRM for ServierBD | *HRM / Multinational Enterprise* *Role: Backend Development*

Developed a custom Human Resource Management system tailored to the organizational workflows, employee lifecycle, attendance, and HR operations of a multinational pharmaceutical company. **[Python, Django, PostgreSQL, HRM]**

Multi-Branch POS & Inventory System | *Retail / Business Management* *Role: Full-Stack Development*

Built a multi-branch point-of-sale and inventory system for managing daily retail billing, barcode stock tracking, inter-branch transfers, and financial operations across multiple locations. **[Python, Django, Next.js, PostgreSQL, MySQL]**

REST API Platforms for Mobile Apps & SaaS Products | *APIs / Mobile & Third-Party Integrations* *Role: Lead Backend Developer*

Architected and delivered multiple REST API backends powering iOS/Android mobile apps and SaaS platforms (including AI healthcare report processing, social media management, and restaurant inventory systems) with full API documentation. **[Django REST Framework, FastAPI, PostgreSQL, OpenAPI/Swagger, Third-Party APIs]**

Websites, Web Portals & E-Commerce Platforms | *Full-Stack Web / E-Commerce* *Role: Full-Stack Development*

Developed and contributed to numerous full-stack web applications — including e-commerce storefronts, multi-vendor portals, discussion forums, blogs, and corporate websites. **[Django, DRF, React.js, Next.js, HTML5, CSS3, JavaScript, Bootstrap]**

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science & Engineering (CSE) — *Green University of Bangladesh* **2020 | CGPA: 3.74 / 4.00**
Diploma in Computer Engineering — *Comilla Polytechnic Institute (BTEB)* **2016 | CGPA: 3.56 / 4.00**
Higher Secondary Certificate (HSC — Business Studies) — *Nazim Uddin Bhuiya University College (Comilla Board).* **2012 | GPA: 4.80 / 5.00**
Secondary School Certificate (SSC — Business Studies) — *Mahajampur High School (Comilla Board)* **2010 | GPA: 4.50 / 5.00**

AWARDS & ACHIEVEMENTS

- **Software Engineer of the Month (3× Recipient)** ***— Esquire Technology Ltd. (2021 – 2022):*** Awarded Software Engineer of the Month three times for exceptional backend architecture, ERP delivery, and engineering leadership.
- **Vice Chancellor's (VC) Certificate — Perfect 4.00 / 4.00 SGPA** ***— Green University of Bangladesh (2019):*** Awarded for achieving highest academic honors in the 6th semester with a perfect 4.00 out of 4.00 GPA.
- **Dean's Certificate — 3.93 / 4.00 SGPA** ***— Green University of Bangladesh (2020):*** Awarded for achieving highest academic honors in the 10th semester with a 3.93 out of 4.00 GPA.
- **Runner-Up Prize — Project Showcasing & Competition** ***— Green University of Bangladesh (2019):*** Recognized as Runner-Up in the university-wide software engineering project showcasing competition.

CONSULTING SERVICES & ADDITIONAL INFORMATION

- **Core Services Offered:** Backend Development · AI Application Development · SaaS Product Development · Business Automation · Database Engineering · Web Development · Custom Software Development (ERP/HRM/POS) · Mobile Apps Development · DevOps Service
- **Languages:** Bengali (Native), English (Professional Working Proficiency)
- **Personal Details:** Nationality: Bangladeshi | Present Address: Dhaka-1217, Bangladesh | Permanent Address: Narayanganj, Dhaka-1440, Bangladesh